

SURVEY

A Survey on Machine Learning Techniques in Smart Grids Based on Wireless Sensor Networks

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ABSTRACT Smart grids (SGs) are crucial to the efficiency and sustainability of modern energy systems. As the world's population continues to increase, so does the need for energy, and traditional energy systems are struggling to keep up. In this context, this study reviews the possibilities of deploying machine learning (ML) on wireless sensor networks (WSNs) in smart grid systems. In several ways, SGs may gain from combining WSNs with ML, including enhance system reliability, sustainability, improve fault detection, and increase energy efficiency. This paper offers an extensive review of pertinent research emphasizing the use of supervised, unsupervised, and reinforcement learning approaches. The evaluation contains 234 peer reviewed articles from highly regarded academic journals and conferences covering the years 2017 through 2024 which depict the effectiveness of supervised techniques on WSNs in the field of SGs. In addition the paper presents set of the most usable datasets in the field of WSNs and SGs, and introduces a comparison between our paper and relevant surveys. The study also analyses the opportunities and challenges related to the application of WSNs and ML in SGs and offers possible research directions. Overall, the study makes it clear that combining WSNs with ML may significantly contribute to the creation of smart grid systems that are more effective, dependable, and sustainable.

INDEX TERMS Wireless sensor networks, smart grids, machine learning, supervised ML, smart sensors.

I. INTRODUCTION

Electric Power System (EPS) is one of the most common types of energy that affects humanity's life, economy, and governmental issues. It is generated in the basic foundation known as the power network or grid. A power network consists of many parts linked and distributed in numerous places [2]. EPS consists of three main stages, firstly, generating stations usually called (power plants). Then a Transmission stage transports Power efficiently from power plants to consumers via transmission lines. The final stage is electricity distribution to consumers. The three stages are depicted in Figure 1.

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The consumer's demand level may be increased over the distributed level or decreased. There is a need to control and manage the power system, load balancing, and automation system. The demand and supply of electrical energy must be balanced, the system operator uses data provided by a higher-level control and data acquisition framework called Supervisory Control and Data Acquisition (SCADA) to observe, manage, and control the generation and transmission system within the Energy Management System (EMS). References [1], [3], [4], [5], and [6] presented other system control approaches for smart automated systems depending on (SG).

An SG is an energy network that continuously monitors the energy flow and adapts to any change in power supply or energy demands. SG uses the smart metering system to

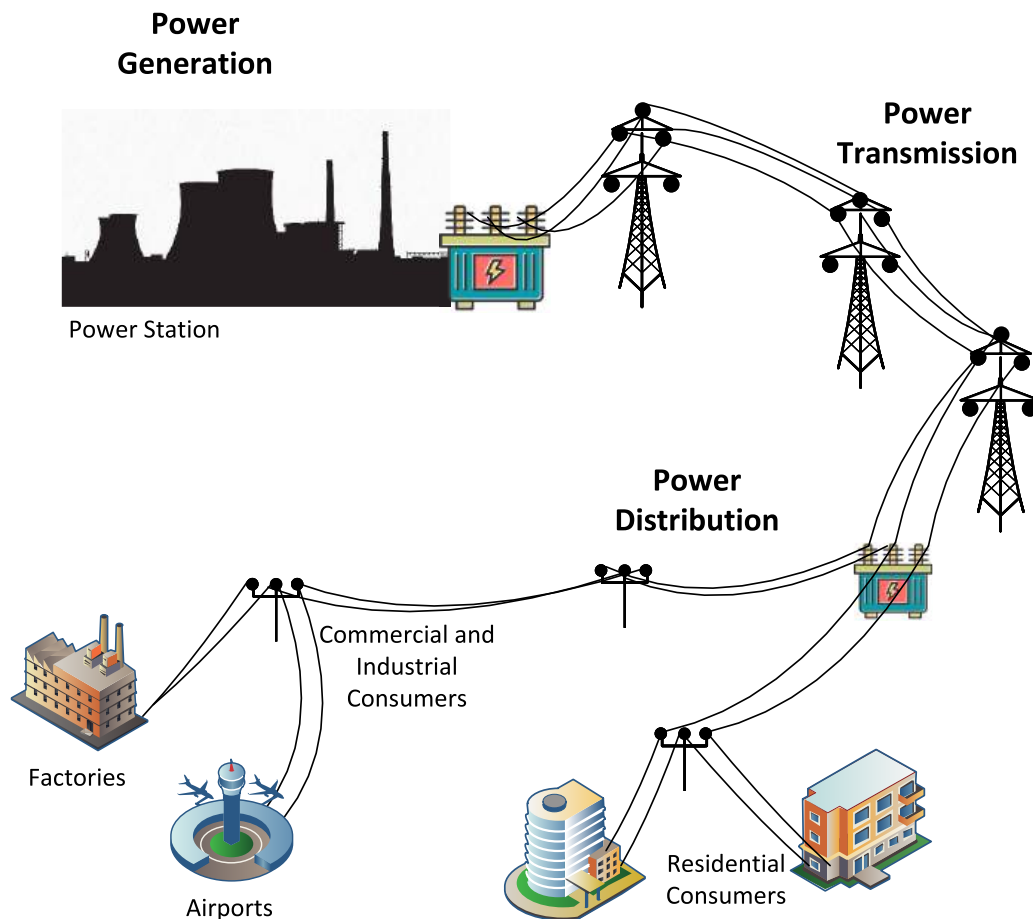


FIGURE 1. Electrical power system.

measure consumer usage by providing real-time data about consumption [7]. WSNs can be used to build an SG network, WSN consists of a set of deploying nodes that monitor several network metrics. SG networks can be used in autonomous systems such as vehicle network monitoring, power plants, power line management, renewable energy adaptation, etc. [8]. ML can help SGs to become more intelligent, adaptive, and efficient, leading to a more sustainable and reliable energy system.

ML has given a huge immense lift to the world of smart sensors network (SSN) [9], [10], [11]. SGs may use ML algorithms for several applications, including load forecasting [12], [13], fault detection and diagnosis [14], energy management [6], [15], [16], [17], and renewable energy integration [18], [20], [22]. ML techniques can improve SSN for optimal resource utilization and network longevity [24], [27], [28], [29].

Due to the diversity of SG research, different researchers are likely to have varying perspectives and focuses when expressing their aspirations for SGs. This study investigates the three main systems of SG from a technical standpoint in accordance with this framework [30].

Previous surveys on SG were done in [31], [32], [33], [34], [35], [36], and [37] reviewed the main idea behind

SGs and many techniques that could be implemented. In [8] Elwakeil et. al. investigated ways to maximize the number of sensing nodes and packet length to extend the lifespan of the SG network. While in [38] Rohanjans et. al. reviewed the existing standardization and gave basic recommendations for the future of SG standards. On the other hand [24], [39], [40], [41] concentrate on using ML techniques in the field of SG. And in [42], Kumar et. al. gave a broad overview of the most useful DL methods for developing reliable, accurate, and safe intelligence SGs infrastructure. But Sahani et al. [43] studied applying ML in intrusion detection for SG computing. And in [44], Qamar et. al. developed two innovative hybrid algorithms: “Delay Bound Reduced k means with ANN” and “Reduced k-means based on ANN” are used for designing a mobile sink path that is effective, fast, and most competent based on rendezvous points. The new model leads to improved energy efficiency and network lifetime. Table 1 lists the abbreviations utilized throughout the paper. A comparison between our survey and set of relevant survey papers in terms of WSNs Application and the key contribution in each paper is listed in Table 2.

The major contributions of the paper are listed as follows:

- The paper presents a comprehensive review of SGs for sustainable energy.

TABLE 1. Table of abbreviations.

Acronym	Description
A/D	Analogue to Digital Converter
AMI	Advanced Metering Infrastructure
ANN	Artificial Neural Network
ARIMA	Auto Regressive Integrated Moving Average
BSM	Binary Sensing Model
CRS	Cognitive radio sensors
DL	Deep Learning
DT	Decision Trees
DTs	Decision Trees
EMS	Energy Management System
EPS	Electric Power System
EQM	Efficient Queuing Multimedia
FCM	Fuzzy-C-Means
HAN	Home Area Networks
HC	Hierarchical clustering
ICA	Independent Component Analysis
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
KNN	k-nearest neighbors
LAN	Local Area Network
ML	Machine Learning
NAN	Neighborhood Area Networks
NB	Naive Bayes
P2P	peer-to-peer
PCA	principal component analysis
PMU	Phasor measurement unit sensors
QoS	Quality of Service
RF	Radio Frequency
RFC	Random forest Classification
RNN	Recurrent Neural Network
SCADA	Supervisory Control and Data Acquisition
SG	Smart Grid
SSN	Smart Sensor Network
SVD	Singular Vector Decomposition
SVM	Support Vector Machine
UMMS	Underground Mining Monitoring System
WAN	Wide Area Networks
Wi-Fi	Wireless Fidelity
WMN	Wireless Mesh Networking
WSN	Wireless Sensor Network

- The paper depicts that ML techniques and WSNs can improve smart grid performance.
- The paper identifies several challenges related to ML applications in SGs.
- The paper assures that SG systems that integrate ML WSNs can benefit from enhanced energy efficiency, security, fault detection, and monitoring.
- The paper asserts the impact of supervised approaches on WSNs for smart grid applications is demonstrated by the study of 234 peer-reviewed articles (from 2017 to 2024).
- The paper presents set of the most usable datasets in the field of WSNs and SG.
- The paper introduces a comparison between our paper and relevant surveys in the field of WSNs and SGs.
- The paper presents future research areas as well as the possible advantages and difficulties of using ML and WSNs in SGs.

The structure of this paper is as follows: Section II provides the construction and operation of SGs, section III

TABLE 2. Comparison of our article with the existing survey papers.

Paper	WSN Application	Key Contributions
[30]	SGs	Smart infrastructure, management, and protection system analysis
[104]	Cyber-Physical Systems	ML security attack analysis, defense mechanisms, and threat modeling
[122]	Smart Cities	ML techniques for WSN-IoT optimization in smart cities in period (2010-2021)
[123]	SGs	ML for cyberattack detection and mitigation
[43]	SGs	ML-based IDS for smart grid security.
[131]	WSNs	ML for WSN optimization and energy efficiency.
[145]	SGs and Demand Response	DL for load forecasting, state estimation, and energy theft detection.
[183]	SGs	Demand response programs, issues, approaches, and future directions.
[163]	WSNs	ML for WSN synchronization, congestion control, and energy harvesting in the timeline (2014–2018).
[130]	WSNs	DL for WSN data processing and feature extraction.
Our Work	WSNs and SGs	Survey of ML techniques for WSNs in SGs, focus on supervised ML, analysis of strengths and weaknesses, most usable datasets in the field of WSNs, comparison between our paper and relevant surveys in the field of WSNs and SGs, future directions, offers a broader and recent scope (2017–2024) and detailed analysis of supervised ML techniques for WSN optimization in SGs.

represents the WSNs concepts and applications, section IV gives background about ML and its techniques in the field of SGs, section V provides the discussion and the survey report, while section VI presents the challenges and research directions and Section VII presents the conclusion and recommendations for further work.

II. SMART GRIDS

An SG is an advanced electrical power distribution network that integrates traditional power grid infrastructure with modern communication and information technology to enable instantaneous monitoring, analysis, control, and optimization of the flow of electricity [45].

SG networks integrate various devices and technologies, such as sensors, meters, and control systems, to monitor and manage energy usage in real time. They also allow for the integration of renewable energy sources, such as wind and solar power, into the network as shown in Figure 2. An improved, safe, and sustainable energy system is the end outcome [46].

SG networks can be divided into three parts:

- Home Area Network (HAN).
- Neighborhood Area Network (NAN).
- Wide Area Network (WAN) [1] as shown in Figure 3.

Consumers may track their energy usage and power bills using in-home display devices thanks to HAN, which

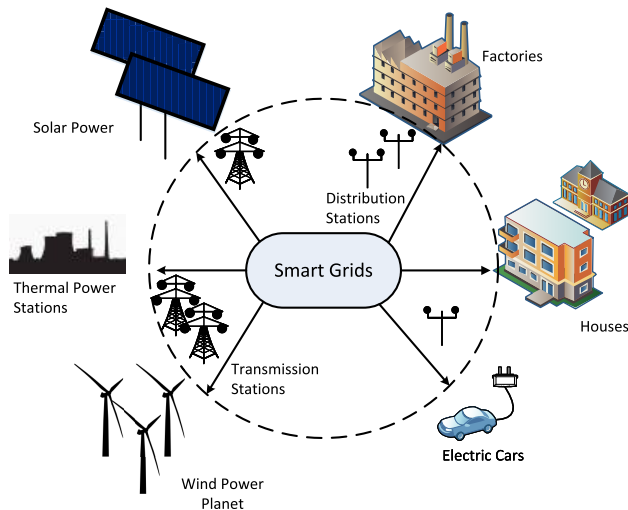


FIGURE 2. Smart grid power system.

establishes a connection link between smart meters, home appliances, and plug-in electric cars. To ensure cost-effectiveness for HAN applications, low-bandwidth connection technologies such as WiFi, Bluetooth, and ZigBee are utilized.

NAN is created between smart meters and data collectors within a specific neighborhood. Data from smart meters is collected and sent to the data concentrator for processing using short-range communication technologies like WiFi and RF mesh.

WAN establishes a connection between the service provider's data centers and data concentrators, enabling high bandwidth and reliable bidirectional connection. The WANs are designed to handle long-distance data transmissions required for monitoring and controlling SG systems. Typically, WANs utilize communication technologies that offer the best coverage at the lowest cost, such as 5G, 4G, LTE, cellular networks (3G and 2G networks), fibers, and energy line communication grids [47].

The Advanced Metering Infrastructure (AMI) is a component of the consumer-side NANs in the SGs as shown in Figure 4. AMI allows utilities to monitor energy consumption in real-time, detect outages, and manage energy distribution more efficiently [48]. Wireless mesh networking (WMN) is a viable option for establishing a NAN due to its self-organizing and self-configuring features, which reduce the need for network maintenance, present reliable service coverage, and improve overall robustness. To establish a NAN using WMN, the IEEE 802.11s standard is one of the open standards available for use [49].

III. WIRELESS SENSOR NETWORK (WSN)

A WSN is a network made up of numerous sensor nodes distributed throughout a designated area, each sensor node can perceive and collect the external environment data. These sensors are typically placed close to each other, within a tens

of meters, to ensure comprehensive coverage. They can be deployed either in bulk or individually. In case of sensor **failure**, additional sensors can be redeployed to replace them, particularly in large or inaccessible fields [50], [51]. With the continuous evolution of sensor technology, leading to sensors that are smaller, more intelligent, and more affordable [52], [53], [54], WSNs have found extensive applications (see Figure 5) across several domains including:

- **Military applications** including observation of the battlefield, reconnaissance of enemy forces and topography, targeting, evaluation of battle damage, and detection and reconnaissance of nuclear, biological, and chemical strikes.
- **Environmental applications** involve forest fire detection, bio-complexity mapping of the environment, flood detection, and precision agriculture.
- **Health applications** include telemonitoring of human physiological data, tracking of doctors and patients within hospitals, and drug administration in medical facilities.
- **In the manufacturing industry**, sensors are employed to monitor the performance of different processes and machine operations. Through data collection, they establish standard baseline levels of operation while also detecting even the slightest fluctuations in performance.
- **Security applications** such as intrusion detection, access control, encryption and authentication, tamper detection, and physical security. They can help detect unauthorized activities, secure data transmission, monitor sensitive areas, and protect critical infrastructure.
- **Home applications** comprise home automation and smart environments.
- **Agricultural domain applications**, the use of a WSN system (wired or wireless) allows for the monitoring and data collection of many different factors, including light, humidity, air and water temperatures, air pressure, sprinkler water flow, soil acidity, and soil moisture.
- **Other commercial applications** incorporate inventory control management, vehicle monitoring, environmental control in office buildings, interactive museums, auto theft detection, and more.

WSN consists of different sensor categories, including thermal, seismic, acoustic, visual, infrared, radar, and magnetic sensors. which can monitor a wide scale of environmental states. Different environmental states are object properties such as temperature, vehicle movement, humidity, pressure, lighting conditions, noise levels, ground conditions, mechanical stresses, presence or absence of objects, size, speed, and orientation. In addition to being used for location and ongoing tracking, sensor nodes recognize and detect events [55].

A. ENVIRONMENTAL SENSORS

According to the environment, sensors are classified into different categories as shown in Figure 6 (i) Terrestrial

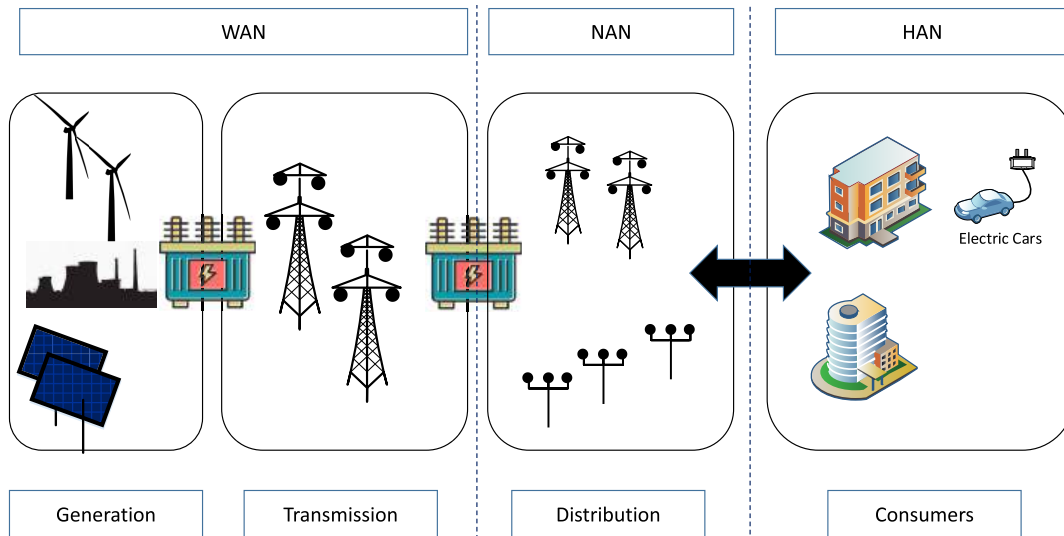


FIGURE 3. Architecture of SG from generation to consumers.

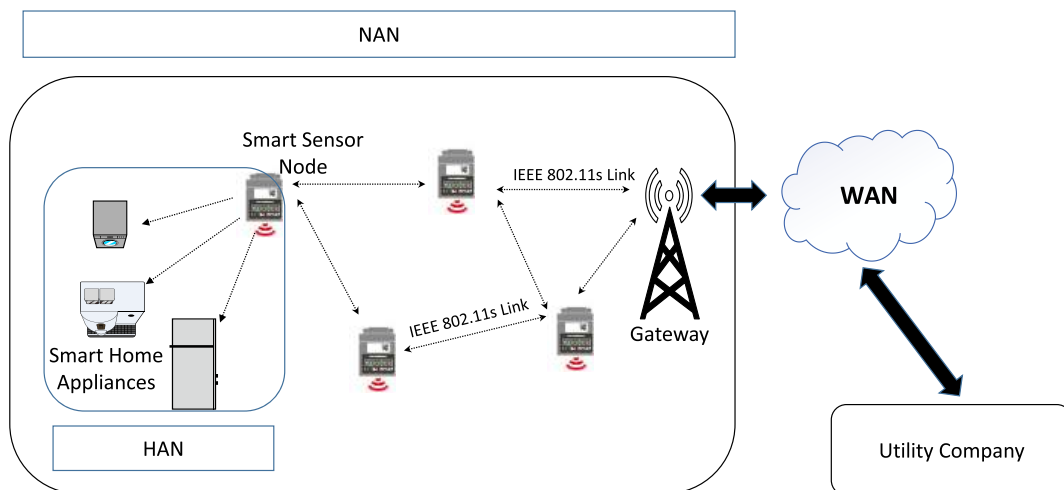


FIGURE 4. IEEE 802.11s-based SG AMI communications.

WSNs. (ii) wireless underwater sensor network. (iii) wireless multimedia sensor network. (iv) underground WSNs.

1) TERRESTRIAL WSNs

Terrestrial WSNs contain hundreds of wireless sensor nodes that can capture events on an ad-hoc or pre-planned basis. For terrestrial WSNs, power consumption and stable communication in dense environments are important. Operating with low-duty cycles, short transmission lengths, minimizing delays, decreasing data redundancy, and effective routing can all help save energy. Dir and Omni are algorithms used to accurately identify objects on Earth [56], [57]. Both algorithms use static or dynamic paths, considering a series of waypoints within the deployment area.

2) UNDERWATER WSNs

In underwater WSNs, sensors are deployed underwater and have special characteristics such as high latency, limited

power, and high error rates [58]. These sensor nodes cost a lot of money to deploy, equip, and maintain. Replacement after use is a difficult task. Rather than an RF signal, an acoustic signal with lower bandwidth and speed is suitable for underwater communication. The protocols used for data routing underwater can be classified as location-based routing protocols or location-free routing methods. The significance of underwater sensor networks has grown recently as a result of the relevance of undersea investigation, marine life study, oil/spill observation, and tsunami and seismic recognition. In an underwater setting, designing a routing protocol is challenging, much like replacing batteries is quite expensive. Another problem is increasing energy efficiency since replacing batteries in water is also a challenge.

3) MULTIMEDIA WSNs

Multimedia WSNs keep an eye on events by collecting data from them in the form of audio, video, and photos. High

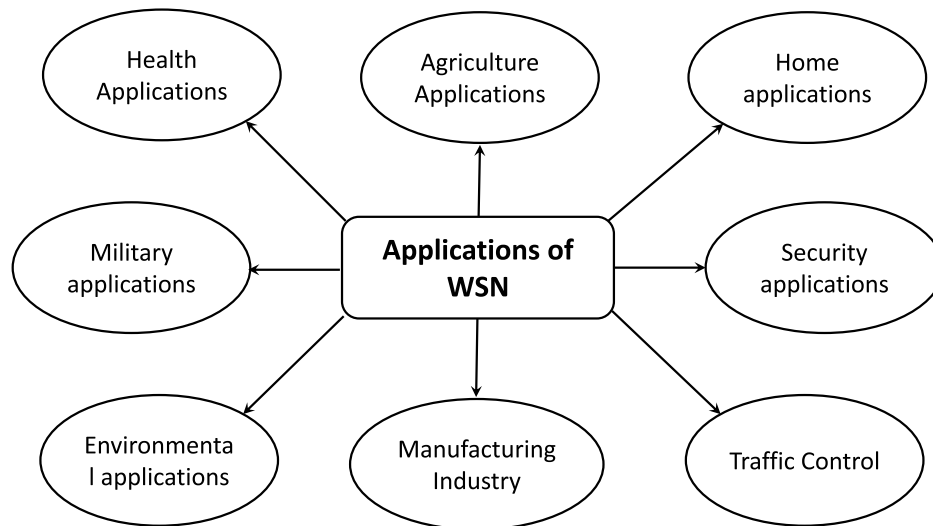


FIGURE 5. Applications of WSN.

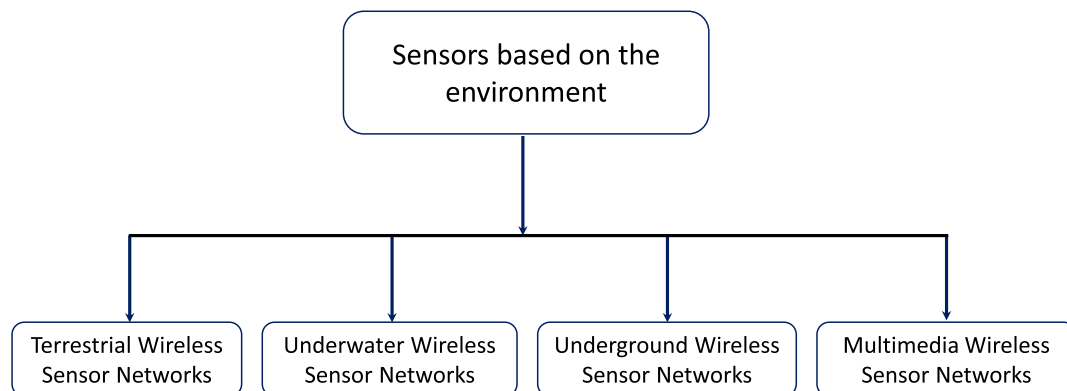


FIGURE 6. Categories of environmental wireless sensors.

quality of services (QoS), high data rates, high bandwidth, massive information processing, most extreme power utilization, and high compression techniques are all features of these networks. Deployment. Video transmission requires high data rates and wide bandwidths, which leads to increased energy consumption. QoS delivery is another difficult problem that plays a key role in reliable content delivery [59]. For video streaming via WSNs, cross-layer techniques and routing protocols have been recommended. A multipath routing protocol called Efficient Queuing Multimedia connection over WSN (EQM) works in concert with the applications layer to efficiently transmit information and videos from various sources.

4) UNDERGROUND WSNs

In underground WSN, sensors are located underground, and reliable communication is costly. Subterranean communication takes place through soils, rocks, and other minerals. A real-time mechanism has been recommended to monitor mining on underground WSNs. Efficient detection

and coverage are achieved using wireless sensor nodes and light sensors. It also detects and identifies damage such as humidity, temperature, and stress vibrations. To wirelessly interact with and monitor miners in underground tunnels, the Underground Mining Monitoring Systems versatile hubs, and reference hubs. To guarantee gearbox reliability and speed, optical sensors are employed. It is equipped with laser diodes, couplers, and a signal transmitter that transforms electrical signals into optical signals. To multiplex light waves, optical multiplexers are employed.

B. NETWORK ARCHITECTURES

In a WSN, a collection of nodes is usually deployed across a specific geographical area, with at least one node functioning as a sink or gateway node. This particular node is responsible for communication with external networks. There are four common WSN topologies as depicted in Figure 7.

1) PEER-TO-PEER TOPOLOGY

In WSN a peer-to-peer (P2P) topology refers to a network architecture where every node in the network can

communicate directly with every other node without the need for a centralized infrastructure or intermediary. This means that each node in the network can both send and receive data and also forward data to other nodes in the network. In a P2P topology, each node is responsible for relaying data between other nodes in the network, creating a self-organizing and decentralized network structure. This can be advantageous in WSNs because it can improve network scalability [60], fault tolerance, and reduce communication latency. However, implementing a P2P topology in WSNs can also be challenging due to limited resources such as processing power, memory, and battery life. Therefore, careful consideration must be given to the design of the network, including routing algorithms [61] and power management strategies, to ensure efficient and effective operation of the WSN.

2) STAR TOPOLOGY

A star topology is a network architecture in which all nodes in the network communicate with a central node called a base station, which acts as a gateway to the outside world. In a star topology, data from sensor nodes is collected by base stations, which send the data to a central location for further processing or analysis. Star topologies are commonly used in WSNs because they are easy to implement and maintain and provide a central control point for the network [1]. It also makes efficient use of resources because nodes can be placed far from each other without the need for additional routing mechanisms. While the star topology also has some drawbacks. If a base station fails, the entire network can fail. Star topologies may also be less scalable than other topologies as the number of supported nodes is limited by base station capacity. WSN star topologies include cluster tree topologies, in which multiple levels of nodes are organized into clusters in which nodes form a self-organizing distributed network structure.

3) CLUSTER TREE TOPOLOGY

In WSNs, cluster tree topology is a hierarchical network architecture that combines the advantages of both star and tree topologies. In this topology, the network is organized into clusters and each cluster has a leader node that communicates with its parent node in a tree structure. WSNs frequently employ cluster tree topologies because they strike a compromise between data aggregation and power efficiency. The cluster head can connect with the network's sensor nodes. This combines the data and moves it up the hierarchy. This saves energy and increases the lifespan of your network by lowering the quantity of data that must be transmitted over long distances. As the failure of one node or cluster head does not always affect the entire network, a cluster tree design can also offer fault tolerance. Additionally, the hierarchical structure of the network can provide scalability as new nodes can be added to the network by creating new clusters. Whereas, the cluster tree topology also has some limitations, such as the potential for congestion at the cluster heads,

which affects network performance. Additionally, network hierarchies can make design and management more complex.

4) MESH TOPOLOGY

A mesh topology in WSNs is a type of network design where each node can communicate with a number of other nodes, resulting in a distributed, self-organizing network structure. In a mesh topology, a node can act as both a source and a relay, sending data to other network nodes so that it may be delivered to its target. Mesh topologies are commonly used in WSNs because they can provide better coverage and fault tolerance than other topologies. Data can travel multiple paths, so it can reach its destination even if some nodes fail or become unavailable. Additionally, mesh topologies are more scalable than other topologies as they can support a large number of nodes without requiring a centralized infrastructure. On the other hand, implementing mesh topologies in WSNs can be challenging due to potential congestion and the need for efficient routing algorithms. Network design, including power management strategies and routing protocols, must be carefully considered to ensure the efficient and effective operation of WSNs.

C. NODE ARCHITECTURE

The node architecture consists of a sensor, microcontroller unit, antenna, and transmitter/receiver systems as shown in Figure 8. The sensor and control units are linked to a battery to provide the necessary power supply voltage. The sensor detects the physical environment and transmits the input to an A/D converter, which converts it into a digital format. Then, the control unit of the microcontroller receives the digital input and utilizes the mechanism stored in the microcontroller unit to the outputs. A sensor is an instrument used to measure a physical quantity and convert it into a signal that is readable by an observer or an instrument. In order to be read on a calibrated glass tube, a mercury thermometer turns the temperature measurement into the expansion and contraction of a liquid.

1) SOME COMMON SENSOR TYPES WITH THEIR APPLICATIONS IN SG

There are many types of sensor nodes used in every aspect of SG and are used in many literature such as:

- **Cognitive radio sensors (CRS):** In [62], the authors suggested that this network can achieve QoS for various types of traffic using a sensor of CRS type.
- **Phasor measurement unit sensors (PMUs):** PMUs can present GPS-synchronized accurate, temporally correlated and much measurements of current and voltage phasors unlike those of traditional sensor [63], whereas in [3], a multi-objective optimization formulation has been used for the optimal placement of PMUs in SG.
- **Thermal/hydro sensors:** They have been utilized in [64] To build a case study structure to analyze a building's thermal performance and determine the most affordable level for a particular building type. The

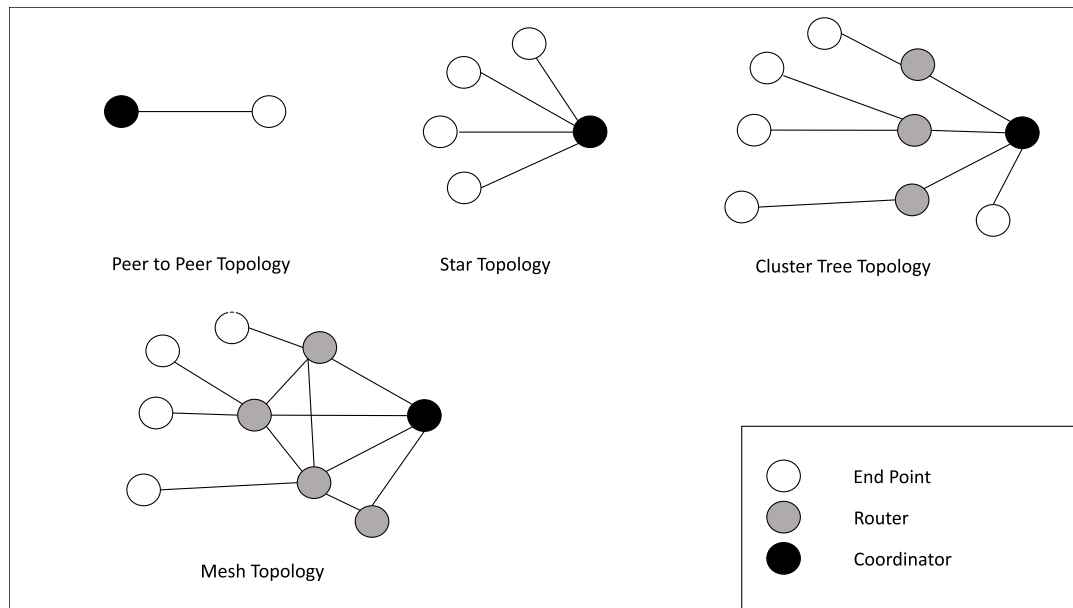


FIGURE 7. Common WSN topologies.

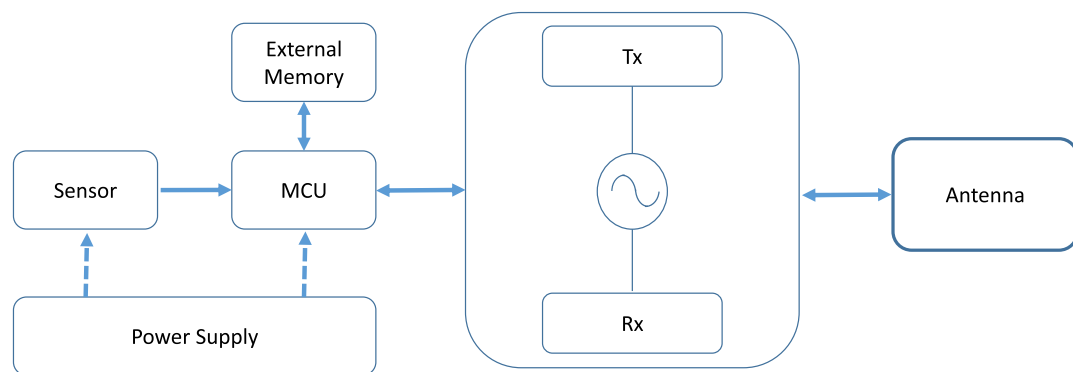


FIGURE 8. Internal node architecture.

usage of thermal sensors in [65] to build a structure for tracking the moment at which a solar system is producing the most electricity. they have been applied in [66] in order To improve the framework for managing heating, cooling, ventilation, and plug-in hybrid electric car charging.

- **Current and voltage sensors:** Current and Voltage sensors are now being applied in [67] to include self-healing components in the distribution network. One of the fundamental traits of SG is its capacity for self-healing.
- **Rechargeable and energy harvesting sensors:** Given that traditional WSNs can be installed in hostile and difficult-to-reach situations, the network's limited battery power may result in shorter lifespans. A high-frequency energy-harvesting sensor is suggested by [68]. The suggested method increased the network's lifespan by allowing the highest priority sensors to receive the most radio frequency energy.

- **Magneto resistive sensors:** A method based on magneto resistive sensors has been suggested in [69] for SG transmission line monitoring. The suggested approach includes computing the source position and current inversely from the magnetic field emitted from the line conductor at ground level.

D. SENSING MODEL

Each sensor node in a WSN has a sensing coverage area based on its sensing range (S_r), which is the range it can monitor as depicted in Figure 9. The network's coverage is the aggregated scope area within WSNs provided from all wake-up nodes, and depending on the radio scope (C_r), the communication distance ensures reliable data transmission within the network. The sensing scope ensures event tracking, while the radio scope ensures the transmission of data. The degree of area monitoring by active nodes is indicated by the coverage degree (C_d). a sensing gap happens when there are dead nodes within a certain area [70]. The network's

lifetime is prolonged by reducing active nodes and achieving the biggest coverage and sensing. The sensing ability of each node in WSN is limited by its range and accuracy, which determines the area it can monitor. There are many factors affecting the sensing model of WSN such as coverage area, lifetime, and redundancy, and are categorized into deterministic or probabilistic sensing models.

1) DETERMINISTIC WSNS' SENSING MODEL

The deterministic or binary sensing Model (BSM) is a sensing model where a sensor node detects events only within its sensing range. Any point not inside the area of interest is not watched. Each node's sensing radius is represented by a uniform circle, abbreviated S_r . At a certain point (p) an event is detected with a probability(s) 1 in the sensing range, while a probability of 0 is assumed if the event occurs outside the sensing range [71]. This is mathematically expressed by:

$$s = \begin{cases} 1, & \text{if } d(i, p) \leq S_r. \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where, S_r represents the sensing range and $d(i, p)$ represents the Euclidean distance [72]. On the other hand, the variable radii circular sensing model is another name for the deterministic sensing model [71]. A subclass of the deterministic sensing model called the binary sensing model is further broken down into perfect binary sensing and imperfect binary sensing models. In the former, if the target is within the sensing range, each node detects it, whereas, latter, the target is found within the inner disc. Despite being straightforward, the binary sensing model does not account for measurement error [73].

2) PROBABILISTIC SENSING MODEL

Although the binary sensing model is straightforward, it does not account for the measurement's ambiguity. It considers various factors that affect the performance of sensing range devices, such as noise, obstacles, and interference. Exponential, polynomial, or staircase distance functions are some examples of the probabilistic sensing models that have been developed to take these elements into account [74]. The probabilistic sensing technique is better for event detection, environment factors, and sensor deployment [73]. It is further divided into several sensing models, including Elfes, Shadowing Fading, Log-normal Shadowing, Rayleigh Fading, and Nakagami-m Fading models [75].

- **Elfes Sensing Model:** It is proposed in [76], where the event detection of a sensor at a distance x is calculated as:

$$p(x) = \begin{cases} 1, & x \leq R_1 \\ e^{-\lambda(x-R_1)^\gamma}, & R_{\max} > x > R_1 \\ 0, & x \geq R_{\max} \end{cases} \quad (2)$$

where R_1 represents the sensing uncertainty in sensor's detect capabilities, λ and γ are various device oriented parameters and $R_{\max}$ defines the maximum sensing

range of the node. Elfes sensing model becomes Boolean sensing model when $R_1 = R_{\max}$. When $R_1 = 0$, $\gamma = 1$, equation (2) becomes equation (3) as given below:

$$p(x) = \begin{cases} e^{-\lambda x}, & R_{\max} > x \geq 0 \\ 0, & x \geq R_{\max} \end{cases} \quad (3)$$

- **The shadowing fading sensing model:** considers the impact of obstacles or other factors on the sensing ability of the node. Due to radio wave propagation, which is closely connected to the idea of shadowing, a node's sensing capacity is not constant in all directions. The model of log-normal shadowing route loss [77], the probability of finding a certain thing at a distance x is calculated as:

$$P_{det}(x) = Q\left(\frac{10m \log_{10}(x/r_s)}{\sigma}\right) \quad (4)$$

where m defines the path loss exponent, sensing radius unfading is represented by r_s and the fading parameter is described by σ .

- **Rayleigh Fading Sensing Model:** This model is used to describe the statistically significant time-correlation of the signal that was received or the individual multi-path envelope component [78]. This fading model is derived from dynamic contexts and a probabilistic viewpoint. It is employed in scattering environments where the received signal may be computed for each site where a receiver can be placed in different locations. No path will prevail at the receiver if there is no line-of-sight path [79]. However, in the presence of a dominant path, a complex Gaussian with a non-zero mean will result. Let x and y represent the in-phase and quadrature components of the Gaussian, respectively. In equation (5), the signal amplitude for the resulting delay is shown.

$$z = \sqrt{(x^2 + y^2)} \quad (5)$$

If these in-phase and quadrature components have variance σ^2 and zero mean, the resulting distribution become Rayleigh [80], which is given as:

$$f(z) = \begin{cases} \frac{z}{\sigma^2} \exp\left(-\frac{z^2}{2\sigma^2}\right), & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases} \quad (6)$$

- **Nakagami-m Fading Model:** It is a mathematical model that illustrates how radio waves weaken as they go across a wireless channel. The variability of the received signal's amplitude brought on by several signal routes is referred to as small-scale fading. The model is expressed mathematically as follows:

$$f(r^2) = \left(\frac{m}{s}\right)^m \frac{r^{2m-2}}{\tau(m)} e^{-mr^2/s} \quad (7)$$

where r^2 refers to the impact of Nakagami-m fading, m is the level of Nakagami-m fading, s is set to 1, and τ

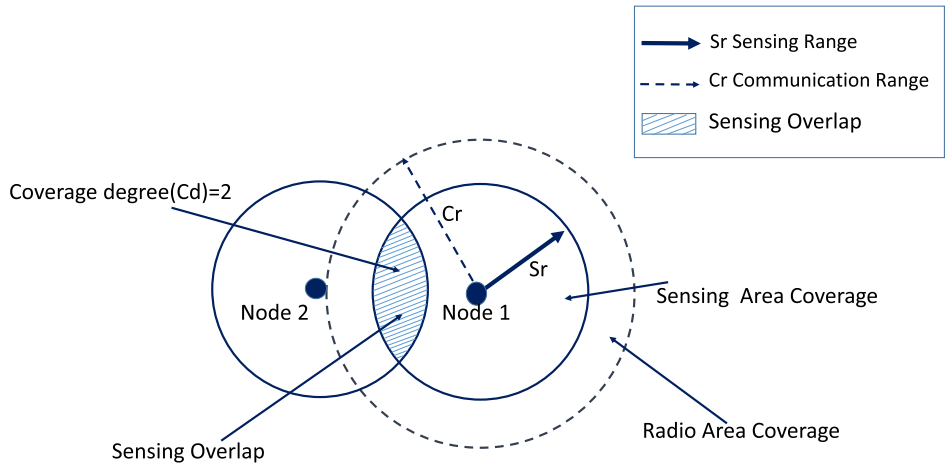


FIGURE 9. Sensing range of a node.

(m) indicates the Gamma function as given by:

$$\tau(m) = \int_0^\infty t^{m-1} \exp(-t) dt \quad (8)$$

IV. MACHINE LEARNING IN SGS BASED ON WSN

In order to construct SG systems based on WSN, ML has grown to be a crucial component. In order to find patterns, spot abnormalities, and forecast future energy demand, these sensors create enormous amounts of data, which is examined by ML algorithms. In addition, ML can be used by SG systems to improve grid stability and resilience while also optimizing energy distribution and reducing energy waste [81]. Moreover, complex demand response strategies can be created with the use of ML, in which consumers are rewarded for lowering their energy use during periods of high demand. As a result, the grid may experience less strain, fewer new power plants may be required, and total energy prices may be reduced [82]. Figure 10 shows the results of our literature review on WSNs in SGs using ML methods. Table 3 summarizes the main datasets used for the discussed ML techniques.

A. SUPERVISED MACHINE LEARNING

A common technique for handling data in ML problems is supervised learning. It predicts outcomes of new inputs through pairs of inputs and outputs that are involved within the training model [23], [24], [25], [26]. Each pair of input and output represents a model f , which comprises x as an input with its corresponding y as a supervised signal ($f : x \rightarrow y$). The model is trained and learned through inputs, a supervised learning algorithm can derive an inferred function that can accurately predict outcomes for new inputs. Supervised learning algorithms have a wide range of applications in SGs using WSN, including localization in [83], [84], and [85], anomaly detection is covered in [86], [87], [88], [89], and [90], quality of service (e.g., [91], [92], [93], [94], [95], [96], synchronization is studied in [97],

TABLE 3. Datasets samples used in related works.

Dataset Name	Reference	Number of Features
WSN-DS	[105]	23
IRLB	[87]	15 possible roles
PQD	[97]	depending on the specific implementation
WSN-BFSF	[101]	18
ESCO	[13]	constantly evolving
KDDCup99,NSL-KDD	[86]	41

and [98], the detection of intrusion and security [99], [100], [101], [102], [103], [104], [105], [106], [107], [108], [109], [110], [111], furthermore integrity of data is discussed in [112], [113], [114], and [115]).

1) REGRESSION

The prediction of target values through supervised methods depends on a set of independent variables [116], [117]. Regarding the type of link between the dependent and independent variables, many regression models are employed. Mathematically, linear regression can be represented as:

$$Y = f(x) + \epsilon \quad (9)$$

where x represents input as independent variable, and Y represents the output as dependent variable, while ϵ represents the error as random value. The regression technique is used to deal with data aggregation [118], [119], [120], localization [83], [85] and energy harvesting in [121] and [122] issues in WSN.

2) CLASSIFICATION

Classification ML plays a crucial role in WSNs and SG applications. In WSNs, classification algorithms are widely used for various tasks such as intrusion detection [123], fault diagnosis, and traffic classification. In SG applications, classification algorithms are used for load forecasting [124], [125], fault detection and diagnosis [126], and renewable

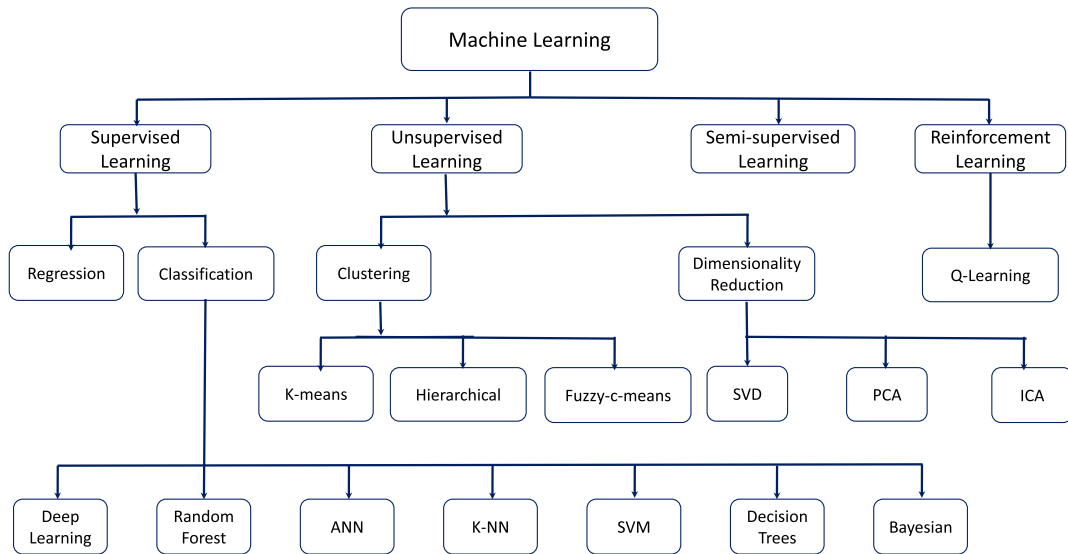


FIGURE 10. Machine learning techniques used in WSN.

energy forecasting [127]. Moreover, classification algorithms are used to classify the customers based on their energy consumption patterns [129] and there are 7 sub-types of classification ML:

- 1) **Decision Trees (DTs):** DTs are supervised ML techniques that may be applied to applications requiring classification and regression. It moves via decision nodes and produces the result through leaf nodes [24]. The mathematical model of DTs model is:

- a) **Tree Structure:** Let T be a decision tree, where N is the set of nodes, E represents the set of edges, each internal node $n \in N$ represents a feature A and each leaf node $l \in N$ represents an output class C or a continuous value.
- b) **Splitting Criteria:** The decision to split nodes is based on certain criteria, such as:
Gini Impurity:

$$Gini(S) = 1 - \sum_{i=1}^C p_i^2 \quad (10)$$

where p_i is the proportion of class i in set S .

Entropy:

$$H(S) = - \sum_{i=1}^C p_i \log_2(p_i) \quad (11)$$

- c) **Information Gain:** Information gain A is obtained from:

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v) \quad (12)$$

- d) **Prediction:** For Classification: The predicted class for a sample x is:

$$\hat{C}(x) = \underset{c \in C}{\operatorname{argmax}} \left(\sum_{x_i \in L} I(x_i = c) \right) \quad (13)$$

where L is the set of samples in the leaf node reached by x .

To create a training model that can recognize the result or class of fresh incoming data, DTs use training data and classification rules. Decision trees have proven to be useful in addressing various design challenges in WSNs, including link reliability identification [130], connectivity [131], anomaly detection [132], [133], and selection of mobile sink path [131]. Over time, researchers have made significant contributions to enhance the effectiveness of decision trees in WSNs.

- 2) **Support Vector Machine SVM:** SVMs are a kind of supervised ML that can be used for classification and regression problems. SVMs are frequently utilized to predict the class of a new data point using a collection of labeled training data. By identifying a hyperplane, SVMs split the data points into two groups. On one side of the hyperplane, all the data points in one class are located while the data points in the other class are located on the other side. A line or plane known as the hyperplane divides the data points in this manner. The support vectors are the data points that are on the hyperplane. Those indicated above are the most important data points for determining the precise location of the hyperplane. SVM basic mathematical

model is:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 \quad (14)$$

$$\text{subject to : } y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1, \quad \forall i \quad (15)$$

where $\mathbf{w}$ represents the weight vector, b is the bias, y_i is the class label for the i th training example (+1 or -1) and $\mathbf{x}_i$ is the feature vector for the i -th training example. And the decision function for classifying a new example $\mathbf{x}$ is given by:

$$f(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x} + b) \quad (16)$$

SVMs are a useful method that may be used to solve a variety of problems. SVMs have been used in WSNs to address a variety of problems, including:

- **Target tracking:** In a WSN, SVMs can be utilized to follow the advancement of targets. This can be achieved via preparing an SVM utilizing a bunch of information that contains the objective's area at different periods [134]. The SVM approach may then be used to forecast the target's future position.
- **Connectivity issues:** In a WSN, SVMs can be utilized to find and address network issues. To do this, an SVM can be prepared utilizing a bunch of information that contains insights concerning the organization's geography and the connection quality between hubs [135]. The SVM may then be used to predict which links will break more frequently and offer solutions for these problems.
- **Routing:** The most efficient routes for data in a WSN may be found using SVMs for routing. An SVM may be trained on a collection of data that includes information about the network architecture, the link quality between nodes, and the energy consumption of each node in order to achieve this. The best data-travel routes may then be predicted using the SVM [136].
- **congestion control:** A WSN's congestion can be managed using SVMs. To do this, an SVM may be trained using a set of data that includes details about network traffic and each node's energy use [137]. SVM may then be used to predict when blockage is likely to occur and to suggest solutions for this problem.
- **Fault detection:** SVMs might be utilized in a WSN to find faults. An SVM that has been trained on a collection of data that contains details on both the network's typical and aberrant behavior may be used to do this. SVM may then be used to forecast the most likely time for a fault to occur [138], [139].
- **Localization:** WSN's nodes may be located using SVMs. To do this, an SVM that has been trained on a set of data that includes node position and signal intensity information may be employed. Next, the

SVM method may be used to estimate where each node will exist [140].

SVMs are a potent tool that can be utilized to address several issues in WSNs, where in [141] by using SVM classification, it was possible to identify the failure data for cluster head aggregation in WSN. However, it is important to note that SVMs are not perfect as SVMs can be sensitive to noise and outliers, and they can be difficult to interpret [142], also [143] Optimized SVM-based congestion control in WSNs based Internet of Things, in addition, [134] studied the effectiveness of SVM in management of distributed renewable energy resources in WSNs, whereas [15] and [144] used SVM based intrusion detection system with reduced input features for advanced metering infrastructure of SGs. On the other hand, before employing the results of SVMs to guide judgments, it is crucial to carefully assess them.

- 3) **Deep Learning (DL):** DL is a subset of ML that makes use of ANNs to learn from data. DL models, which draw their inspiration from the human brain, may learn to solve issues by examining data and finding patterns [145]. The basic mathematical model of DL is: For a given input $\mathbf{x}$, the output of the network can be computed using the following formula:

For the (l)-th layer, the output ($\mathbf{a}^{(l)}$) is computed as:

$$\mathbf{z}^{(l)} = \mathbf{W}^{(l)} \mathbf{a}^{(l-1)} + \mathbf{b}^{(l)} \quad (17)$$

$$\mathbf{a}^{(l)} = \sigma(\mathbf{z}^{(l)}) \quad (18)$$

where $\mathbf{W}^{(l)}$ is the weight matrix for layer l , $\mathbf{b}^{(l)}$ is the bias vector for layer l , $\mathbf{a}^{(l-1)}$ is the activation from the previous layer, and σ is an activation function (e.g., ReLU, sigmoid, or tanh).

Loss Function:

$$L(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_i y_i \log(\hat{y}_i) \quad (19)$$

where $\mathbf{y}$ represents the true label vector and $\hat{\mathbf{y}}$ is the predicted output from the network.

Backpropagation: The weights and biases are updated using the next equations

$$\mathbf{W}^{(l)} \leftarrow \mathbf{W}^{(l)} - \eta \frac{\partial L}{\partial \mathbf{W}^{(l)}} \quad (20)$$

$$\mathbf{b}^{(l)} \leftarrow \mathbf{b}^{(l)} - \eta \frac{\partial L}{\partial \mathbf{b}^{(l)}} \quad (21)$$

where η is the learning rate.

A wide range of applications, such as chatbots, virtual assistants, and self-driving automobiles, have been created using DL. A range of issues in WSNs in the SG field are being addressed through the usage of DL. In the SGs field, DL can be used to:

- **Optimize power flow:** The current state of the grid can be examined using DL models, and the best routes for power flow can be found. This may

contribute to lowering losses and enhancing grid dependability [146].

- Estimate the quality of data: Data accuracy can be increased by using DL models to recognize and eliminate noise from the data [147].
- Harvest energy: The network's energy utilization can be optimized by using DL models to predict when energy will be required [148].
- Route data: DL models can be used to determine the most efficient paths for data to go, which can increase the network's effectiveness [149].
- Improve grid security: Data analysis and the detection of potential security vulnerabilities can be done using DL models [150] and [151]. The grid's security can be strengthened using this knowledge, and hacks can be avoided.

DL is an effective method that may be applied to solve several problems in WSNs, such as: energy-efficient routing, estimate the quality of data and anomaly detection (improve WSN security). And specific problems in SGs, such as: Optimize power flow, harvest energy and improve grid security. It is expected that DL will be utilized to address more problems in SGs as DL technology advances, such as real-time decision making: DL can enable real-time decision-making in WSNs by processing sensor data in real-time.

- 4) **Random Forest (RFC):** Random forest is a supervised ML technique that may be used for both classification and regression tasks, and it is more frequently utilized for classification issues [152] and [153]. The technique is based on the idea that regular forests are stronger when there are more trees present. Similar to this, the random forest algorithm builds numerous decision trees from data samples and then uses voting to forecast the best outcome. Here is the mathematical model of RFC:

Let X be the feature matrix and Y be the target variable. Let X_i be the i -th feature vector in X with $i = 1, 2, \dots, n$, and Y_i be the corresponding target value.

For $k = 1, 2, \dots, K$, where K is the number of trees in the forest:

- a) Sample a bootstrap sample X_k^* of size N from the training data X .
- b) Grow a decision tree T_k using the sample X_k^* :
 - i) At each node of the tree, randomly select a subset of features to consider for splitting.
 - ii) Split the node based on the feature that provides the best split according to a chosen criterion (e.g., Gini impurity, information gain).
 - iii) Repeat the splitting process until a stopping criterion is met (e.g., maximum tree depth, minimum node samples).

The prediction of the Random Forest model is based on aggregating the predictions of individual trees:

$$\hat{Y} = \frac{1}{K} \sum_{k=1}^K T_k(X) \quad (22)$$

RFC algorithm is effective at managing big and diverse amounts of data thanks to this method. The random forest algorithm in WSNs can address a number of problems, including coverage [154] and MAC protocol [155].

- 5) **Artificial Neural Network (ANN):** ANN is a supervised ML technique that resembles a biological neuron network for data classification [156]. It is made up of a network of interconnected nodes or units known as artificial neurons, which function similarly to synapses in the human brain in that they may convey signals to one another. There is an input layer followed by a hidden layer or sometimes not, and finally output layer forms ANN. The basic mathematical model of ANN is:
1. Neurons in a Layer: For a neuron j in layer l , the output $a_j^{(l)}$ is computed as:

$$a_j^{(l)} = \sigma(z_j^{(l)}) \quad (23)$$

where $z_j^{(l)}$ is the weighted sum of inputs from the previous layer:

$$z_j^{(l)} = \sum_i w_{ji}^{(l)} a_i^{(l-1)} + b_j^{(l)} \quad (24)$$

where $w_{ji}^{(l)}$ represents the weight from neuron i in layer $l-1$ to neuron j in layer l , $b_j^{(l)}$ is the bias for neuron j in layer l and σ is the activation function (e.g., ReLU, sigmoid, or tanh).

2. Forward Pass: The output of a given input $\mathbf{x}$ is computed by propagating the input through the layers:
 - For the input layer (layer 0):

$$a_i^{(0)} = x_i \quad \forall i \quad (25)$$

For each subsequent layer, compute z and a until the output layer.

3. Loss Function:

$$L(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_i y_i \log(\hat{y}_i) \quad (26)$$

where $\mathbf{y}$ represents the true label vector and $\hat{\mathbf{y}}$ is the predicted output from the network.

4. Backpropagation: To update the weights and biases

$$w_{ji}^{(l)} \leftarrow w_{ji}^{(l)} - \eta \frac{\partial L}{\partial w_{ji}^{(l)}} \quad (27)$$

$$b_j^{(l)} \leftarrow b_j^{(l)} - \eta \frac{\partial L}{\partial b_j^{(l)}} \quad (28)$$

where η represents the learning rate.

ANN has been used in research related to WSNs and SGs, despite their need for high computing power. ANN is particularly useful in detecting faulty sensor nodes [157], maximizing the network lifetime [158], performing data aggregation [159], SG security [160] and congestion control [161].

- 6) **Naïve Bayes (NB)**: NB or Bayesian is a supervised ML technique that utilizes a statistical learning technique for classification. Both binary (two-class) and multi-class classification issues may be solved using it as a classification procedure. The technique makes it easier to determine the probability for each hypothesis given a collection of inputs, making its computations easily traceable. NB basic mathematical model is:

1. Bayes' Theorem: The probability of a class C given a feature vector $\mathbf{x} = (x_1, x_2, \dots, x_n)$ is given by:

$$P(C | \mathbf{x}) = \frac{P(\mathbf{x} | C)P(C)}{P(\mathbf{x})} \quad (29)$$

2. Naïve Assumption: Assuming that the features are independent given the class, we can simplify $P(\mathbf{x} | C)$:

$$P(\mathbf{x} | C) = \prod_{i=1}^n P(x_i | C) \quad (30)$$

3. Final Classification Rule: To classify a new instance, we choose the class C that maximizes the posterior probability:

$$\hat{C} = \arg \max_C P(C | \mathbf{x}) = \arg \max_C (P(\mathbf{x} | C)P(C)) \quad (31)$$

4. Putting It All Together: The classification can thus be expressed as:

$$\hat{C} = \arg \max_C \left(P(C) \prod_{i=1}^n P(x_i | C) \right) \quad (32)$$

In WSNs, Bayesian inference is commonly used to assess event consistency (h) using incomplete data sets (D) by leveraging prior knowledge about the environment [161] and in improving time synchronization [162].

- 7) **k-nearest neighbors (KNN)**: The KNN algorithm is a supervised ML technique that can handle regression and classification issues, as the size of the dataset used steadily affects the complexity of the algorithm. However, the prediction procedure might get longer as the bulk of the data grows. To calculate the distance between two input locations, KNN uses a variety of distance functions, such as the Euclidean distance [163]. Additionally, it can decrease dimensionality and locate any missing values in the feature space [164]. KNN basic mathematical model is:

1. Distance Metric: first calculate the distance between the query point $\mathbf{x}$ and all points in the training dataset.

$$d(\mathbf{x}, \mathbf{x}_i) = \sqrt{\sum_{j=1}^m (x_j - x_{ij})^2} \quad (33)$$

where $\mathbf{x}_i$ represents the i -th training example and m is the number of features.

2. Selecting Neighbors: next step is to select the k nearest neighbors.

$$\mathcal{N}_k(\mathbf{x}) = \{\mathbf{x}_{i_1}, \mathbf{x}_{i_2}, \dots, \mathbf{x}_{i_k}\} \quad (34)$$

where $\mathcal{N}_k(\mathbf{x})$ represents the set of the k neighbors closest to $\mathbf{x}$.

3. Voting for Classification: For classification, the predicted class $\hat{y}$ of the query point $\mathbf{x}$ is determined by majority voting among the k nearest neighbors:

$$\hat{y} = \text{mode}(y_{i_1}, y_{i_2}, \dots, y_{i_k}) \quad (35)$$

where y_{i_j} is the class label of the j -th neighbor.

Due to its simplicity, KNN is particularly useful for query processing tasks in WSNs. In WSNs and SGs, KNN can be used to solve problems such as mobile sink, anomaly, and fault detection [165], as well as data aggregation [75].

B. UNSUPERVISED ML

The unsupervised algorithm is an ML model that is completed in the absence of supervision. Instead, it allows the model to discover hidden patterns and information without the need for labeled datasets. Unsupervised learning is mainly used to handle unlabeled information. This approach has made significant contributions to resolving various issues in WSNs and SG field, such as routing [39], [166], [167], connectivity problems [122], [168], data aggregation [166], [169], quality of service [170] and anomaly detection [171], [172], [173]. There are two main branches of unsupervised learning:

- Clustering.
- Dimensionality reduction.

1) CLUSTERING

Unsupervised clustering ML is classified into:

- 1) **K-mean**: By using iterative distance-based measurements, the k-means clustering approach separates a given dataset into 'k' distinct, beforehand determined, and mutually exclusive groups. The algorithm aims to achieve high intra-cluster similarity and low inter-cluster similarity [174]. K-means clustering may be employed in SGs employing WSNs to identify the ideal cluster heads for data routing to the base station [39], while [170] studied how to improve QoS in AMI applications in SGs, and [168] suggested a novel hybrid clustering model for WSN using k-mean clustering algorithm.
- 2) **Hierarchical clustering (HC)**: HC is an unsupervised ML algorithm that groups unlabeled data points

based on their similar characteristics. In agglomerative hierarchical algorithms, each data point is first handled as a separate cluster, and then successive pairings of clusters are merged or aggregated. In contrast, in divisive hierarchical algorithms, the data points are seen as a single enormous cluster, which is then divided into numerous smaller clusters [175]. In WSNs, HC can efficiently address issues such as synchronization as [176] presented scheduling policy in cluster-based WSNs, while [177] improved data aggregation by HC, and energy harvesting is implemented efficiently by [178].

- 3) **Fuzzy-C-Means (FCM):** Bezdek created FCM clustering, sometimes referred to as soft computing, in 1981. Based on how similar the observations are, it uses the concept of fuzzy sets to divide them into either one or more clusters. Fuzzy c-means identifies similar clusters based on intensity, distance, or connectivity and iterates on the clusters until an optimal center is found. FCM is particularly useful for datasets with overlap and outperforms k-means clustering in such scenarios. FCM's time complexity, however, is often greater than that of other approaches and depends on a variety of elements, including the number of clusters, data points, dimensions, and iterations. In SG and WSNs, FCM can solve various issues such as localization [84], [179], connectivity [180], and mobile sink [181].

2) DIMENTIONALITY REDUCTION

It is an ML technique used to lessen the number of input features in a dataset while retaining the most important information [182]. This can lessen overfitting, increase training speed, and enhance the effectiveness and accuracy of ML models. Feature selection and feature extraction are the two major kinds of dimensionality reduction. While feature extraction entails developing new features that capture the most crucial data in the original features, feature selection entails choosing a subset of the original features depending on their significance.

In WSNs and SGs, dimensionality reduction plays the function of reducing the complexity of the data while maintaining the crucial information. As a result, energy usage in WSNs may be decreased, data analysis can be done more quickly and accurately, and demand forecasting models in SGs can be improved.

Dimensionality reduction strategies in WSNs can aid in reducing the volume of data that must be transferred, extending the life of the network, and lowering energy use. Dimensionality reduction can also enhance the precision of data analysis and lower the danger of overfitting by deleting unnecessary or duplicate features. While in SGs, dimensionality reduction can help reduce the computational complexity of demand forecasting models, which can improve their speed and accuracy [183]. By identifying the most important

features in the data and reducing the dimensionality of the input space, dimensionality reduction techniques can enhance the overall efficiency and reliability of the SG system. The Common approaches to dimensionality reduction include:

- 1) **Singular Vector Decomposition (SVD):** SVD is an unsupervised ML algorithm that is commonly used as a data reduction method. In WSNs and SGs, SVD can efficiently address routing issues related to the communication of data between sensor nodes [184]. Additionally, SVD can be used to address data aggregation-related issues [169], [185]. Additionally, SVD ensures the data's ideal low-rank representation, which can assist data analysis to become more accurate and quick [186].
- 2) **principal component analysis (PCA):** A multivariate analysis feature extraction technique for reducing dimensionality is PCA. PCA's main objective is to reduce the size of a dataset that contains a large number of features that are either weakly or highly related to one another. Dimensionality reduction becomes necessary since the sensor nodes of WSNs conduct ongoing surveillance and transmit massive amounts of data [187]. PCA can assist reduce data dimensionality at the sensor or cluster head level, which lowers transmission costs. Several WSN methods, including localization [188], fault detection [189], and target tracking [190], [191] have also adopted the PCA technique.
- 3) **Independent Component Analysis (ICA):** ICA is an unsupervised ML methodology that makes use of mathematical and statistical techniques to identify potential hidden characteristics in collections of random data, signals, or measurements [192]. ICA is a generative model that uses multivariate information that is frequently used with large sample data sets [193]. ICA has been applied to analyze data from various fields, including corporate information, digital photographs [194], psychometric assessments, web contents, and social networks [195] and An enhanced multiplicative non-negative independent component analysis for detecting multi-speaker speech activity [196].

C. SEMI-SUPERVISED LEARNING

Semi-supervised ML concentrates on labeling the unlabeled data in the training phase and forecasting labels for upcoming datasets during the testing phase. This method is frequently used in real-time applications for things like audio analysis, natural language processing, spam filtering, and web content classification. In SGs and WSNs, semi-supervised learning can be used to resolve issues related to localization [197], [198], fault detection [199], while [200] studied To find power fraud in the distribution networks, a semi-supervised learning approach with attention guidance is used.

TABLE 4. Summary of literature survey of ML techniques in SGs.

SG problem	References	ML technique used	Our remarks
Load forecasting	[12], [13], [124], [125], [206]	DT, SVM, DL and RFC	more studies on Reinforcement ML must be done.
Fault detection and diagnosis	[14], [126], [133], [138], [189], [207], [208]	Reinforcement, Fuzzy, SVM, DL	more studies is needed to reduce fault tolerance.
Traffic classification	[60], [137], [209]–[211]	SVM, FCM, DT, RML	Combination of techniques will get better results.
Energy management	[16], [210], [212]–[219]	RML, Q-Learning, DL, CNN, SVM and NB	More ML techniques will lead to stable energy management.
Renewable energy integration	[8], [18]–[22], [127], [128]	DL, kNN, Regression ML, RFC	As a result of alternatives of renewable energies, the combination of techniques will get better results.
Advanced metering infrastructure	[15], [49], [144], [220], [221]	SVM, RML, kNN, DL	Needs more studies due to the state of art metering systems.

D. REINFORCEMENT LEARNING (RL)

The RL technique for SGs based on WSNs uses a reward system to encourage learning through interacting with the environment. In this approach, a sensor node learns to take optimal actions to maximize its benefits over time, based on experience [201]. Reinforcement learning algorithms can efficiently address routing problems in WSNs, where each node tries to select actions that are expected to increase its long-term benefits [202]. In [203] the sensor node updated the reward it receives according to the activities it took at a certain state using the RL. Q-learning is a type of RL algorithm where each agent interacts with the environment and generates a sequence of observations, such as state, action, and rewards that guide its learning process [204], so [205] expected an important role for RF in decision making under unknown network condition.

V. DISCUSSION

ML techniques are the most advanced approaches used to solve WSNs issues (e.g., energy management, locality, security) in the SGs domains. Supervised ML, unsupervised ML, semi-supervised ML, and Reinforcement ML all can be used to resolve these issues. Over the last few decades, numerous scholars have applied ML algorithms in this subject in diverse ways. In [43], the authors focused on ML-based IDS for SGs from the security view, in [104], the authors highlighted ML security attack analysis, defense mechanisms, and threat modeling only, in [145], the authors used DL for load forecasting, state estimation, and energy theft detection, but in [183], the authors presented demand response programs, issues, approaches, and future directions

TABLE 5. WSN issues with ML-based solutions.

WSN issues	References	ML technique used	Solutions remark
Node coverage and connectivity	[154], [222], [223]	ANN, DT and RFC	Categorization of the WSN's connected and failing nodes
Node Localization	[144], [179], [183], [192], [198], [202], [224], [225]	FCM, FCM, PCA, Regression ML, semi-supervised support vector regression	Range and distance estimation in an efficient manner.
MAC layer	[226], [227]	SVM, DTs, ANN	Efficient channel assignment.
Event monitoring and target detection	[135], [190], [191], [228]	Regression, SVM, DL	Efficient monitoring and multiple target detection.
Node query processing	[229]–[231]	KNN, DL	Handshake for data transfer and node Beacon Sending.
Routing issues	[33], [140], [167], [184], [188]	SVM, Random Forest, DTs, RL	Optimal routing path prediction.
Energy-harvesting	[121], [178], [232]–[234]	DL, SVM, HC, Regression ML	Energy management for improving (prolonging) network lifetime.

for SGs, but in [163], the authors used ML for WSNs synchronization, congestion control, and energy harvesting in the timeline (2014–2018), while our work presented a survey of ML techniques for WSNs in SGs, focus on supervised ML (analysis of strengths and weaknesses), most usable datasets in the field of WSNs, comparison between our paper and relevant surveys in the field of WSNs and SGs, future directions, offers a broader and recent scope (2017–2024) and detailed analysis of supervised ML techniques for WSN optimization in SGs. Concerning the last 8 years (from 2017 to 2024) as shown in Figure 11, supervised ML is the most widely used as depicted in pie char in Figure 12, especially SVM, ANN, and DL the widely used to solve the problems of SGs, and our remarks about each technique used as represented in Table 4.

Table 5 reveals the ML techniques used to solve WSN issues. The top priorities are gathering sensing data and successfully transmitting it to the base station or sink node. If data is sent across an incorrect channel, the nodes' energy level may decrease, potentially leading to a network failure. Therefore, it is essential to specify each node's proper functionality and use appropriate techniques for data sensing and processing. Supervised ML works by automatically training the system based on reference datasets to provide accurate results.

Node location, node coverage, and connection, routing layer, MAC layer, event tracking, gathering data, collecting energy, and node processing of queries are just a few of the significant problems that ML can solve for WSNs.

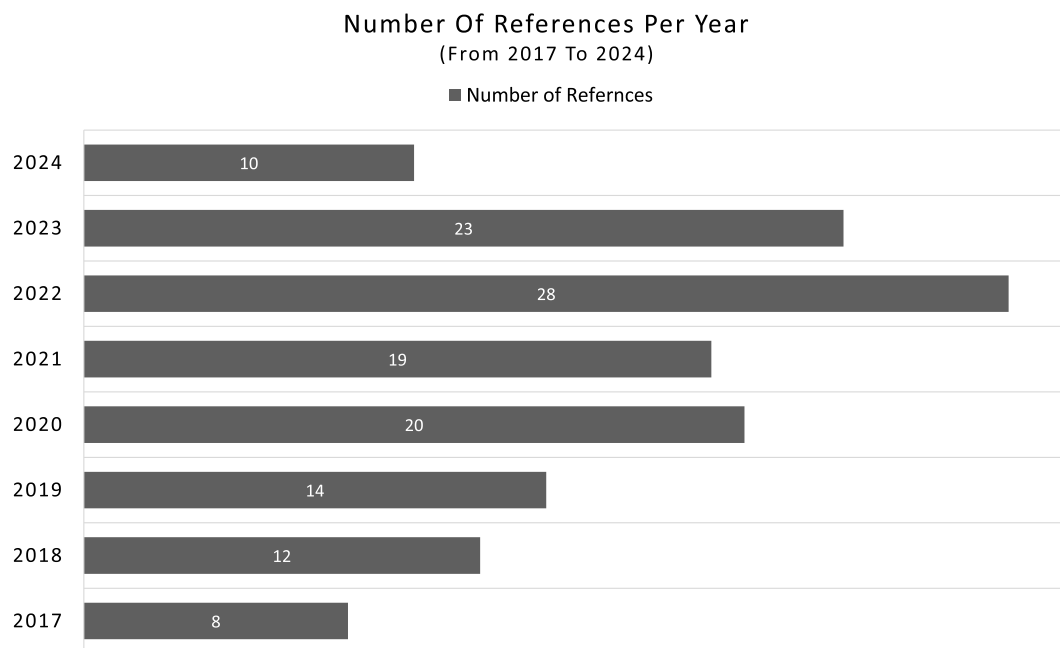


FIGURE 11. Chart number of references per year (From 2017 to 2024) Used in our Survey.

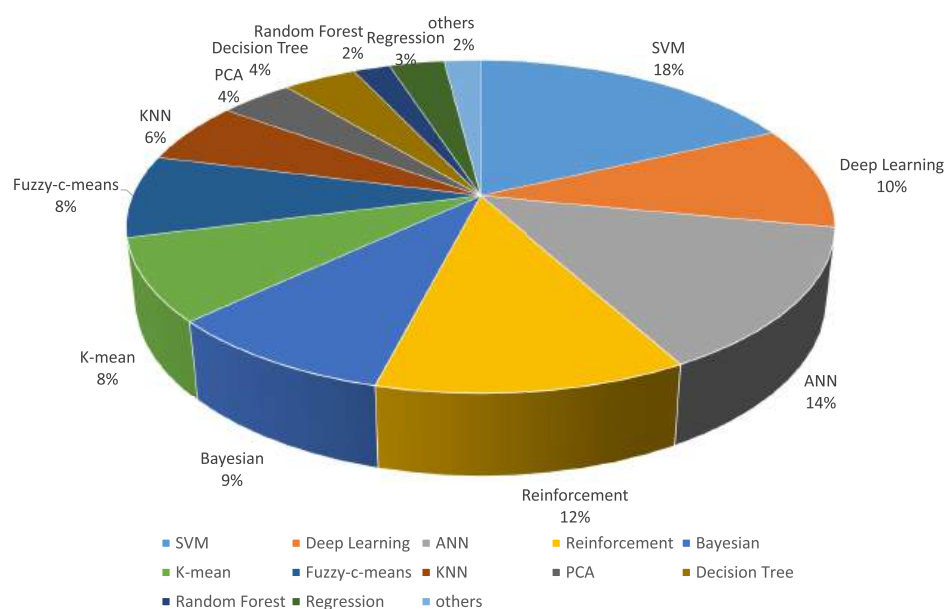


FIGURE 12. Chart depicting the use of ML techniques in WSNs.

So supervised ML techniques are increasingly used in advanced applications, especially with the rapid growth of SGs in the last decade. This trend is expected to continue, highlighting the significant potential for supervised ML techniques in this field.

VI. CHALLENGES AND RESEARCH DIRECTIONS

ML techniques can be used to address a variety of issues in the field of SGs based on WSN. The following is a list of the principal problems that ML can solve:

- **Anomaly Detection:** Anomalies in sensor networks can cause transmission delays, communication overhead, and other issues that can negatively impact the network's lifetime. Anomalies are frequently caused by application-driven problems and might affect the WSNs in many ways. There is a lot of research potential in the use of supervised ML approaches to locate and handle certain abnormalities.
- **Routing:** It may be necessary to gather data from several sources and route it to various destinations in real-world

and practical WSN applications. However, a large possibility of packet collisions might emerge from this. As a result, cooperative WSNs' routing techniques that prevent collisions are necessary. Additionally, for networks whose nodes often change places, a more dynamic routing system can be created.

- **Coverage and Connectivity:** In WSNs especially for SGs, A minimum number of nodes should cover the largest possible area. Coverage gaps can occasionally be produced by the random placement of nodes, though. The network lifetime can be greatly increased if supervised ML techniques are able to forecast such circumstances.
- **Data Aggregation:** Multiple sensor nodes sense the same data and deliver it to the sink in some smart grid applications of WSN. However, handling this data at gateways can be challenging. Therefore, there is a need to summarize and aggregate useful data from multiple sources, which can remove data redundancy and improve the network's lifetime. A more sophisticated machine learning strategy based on data aggregation can be created in order to further enhance this process.
- **Energy Harvesting:** Energy is a significant constraint in large-scale WSNs. power harvesting involves deriving power from alternate sources, such as the environment, and using it to power the network. Even though the majority of WSN protocols now in use are energy-efficient, ML techniques may be utilized to study existing protocols and boost their energy efficiency.
- **Nodes Deployment:** Nodes are organized in a three-dimensional space in several WSNs applications. It may not produce the best results to randomly arrange nodes in this space using the two-dimensional protocols that are currently in use. So, for effective path planning, utilizing a path-learning technique can be quite helpful. There is a lot of room for research in this area.
- **Quality of service issues:** In WSNs, QoS criteria are frequently application-based and must satisfy both the needs of the user and the application. It might be difficult to define QoS for a particular demand and application. In order to solve this problem, cross-layer protocols that can successfully manage QoS in heterogeneous WSN and SG settings can be designed using ML approaches.
- **Target detection and event tracking:** The current use of WSNs in SGs as a component of smart cities strongly relies on target recognition and event monitoring. The two biggest problems in this situation are intrusion detection and traffic monitoring. To address these challenges, supervised ML techniques, such as the Bayesian method, can be very useful.
- **Congestion Prevention and Control:** When it comes to environments that are changing dynamically, conventional congestion control and prevention procedures for WSNs are constrained. To successfully address these changes, congestion control methods should be self-adapting. As a solution to this problem, ML techniques

can be used to improve protocols that can directly gather and analyze sensed data.

VII. CONCLUSION

WSNs and ML techniques in SGs have the potential to revolutionize the energy sector by improving efficiency, sustainability, and reliability. This research focused on the basic ML techniques that may be used to various WSN challenges, mainly in the SG domains. The study shows how flexible ML algorithms are and how a single ML technique can address several WSN issues. This paper offered an extensive review of pertinent research with an emphasis on the use of supervised, unsupervised, and reinforcement learning approaches. In addition the paper presented set of the most usable datasets in the field of WSNs and SGs, and introduced a comparison between our paper and relevant surveys. The evaluation contains 234 articles from highly regarded academic journals and conferences covering the years from 2017 to 2024 that depicted the effectiveness of supervised techniques in the field of SGs, on the other hand, we have pointed to the opportunities and challenges related to applying ML in SGs and offered possible research directions that make SG systems more effective, dependable, and sustainable. In the future work, we aim to present a study containing the pros and cons of other techniques such as Auto Regressive Integrated Moving Average (ARIMA) model compared to the ML ones.

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